

A Computational Method to Generate a Family of Extreme-Value Volatility Estimators

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Abstract

A framework is developed to construct estimators of volatility derived from extreme prices of a security, such as high, low, opening, and closing prices, using null space decomposition. This framework not only reproduces the classical estimators, such as the [Parkinson \(1980\)](#) and [Garman and Klass \(1980\)](#) estimators, but also improves the efficiency of some of the classical estimators, including the [Parkinson \(1980\)](#) estimator. The estimator that utilizes only high and low prices, as derived from this method, achieves a 17% reduction in variance compared to [Parkinson \(1980\)](#), which also utilizes high and low prices. Extension of this framework is possible to cases where prices are observed to drift and discontinuities in prices exist, due to the close of the trading day.

Keywords: Volatility estimation, Extreme values, High-low-open-close, Null space decomposition

JEL Codes: C13, C58, C65, G12

1 Introduction

Volatility of security price returns is not directly observable, but it drives almost every decision in finance. As it is a latent variable, it must be inferred from available price data, either by modeling the latent state or by constructing nonparametric proxies from observed prices. When high-frequency data is available, summing intraday squared returns (“realized volatility”) reasonably estimates daily volatility, but they require careful treatment of market microstructure noise [Alizadeh et al. \(2002\)](#). Alternatively, within the class of daily volatility measures, extreme value (range-based) estimators exploit information in the day’s high and low, often together with the open and close (OHLC), to deliver substantial efficiency gains over close-to-close volatility while remaining data light.

Historically, the literature has advanced several influential OHLC estimators. For example, [Parkinson \(1980\)](#) introduced the “extreme value method”, showing that the high-low range under no drift geometric Brownian motion yields an estimator that is several times more efficient than the close-to-close benchmark. [Garman and Klass \(1980\)](#) combined OHLC in a quadratic form that further reduces the estimator variance under the same benchmark. To address a key limitation of earlier zero-drift estimators [Rogers and Satchell \(1991\)](#) introduced an OHLC estimator that is unbiased for any drift in intraday prices. They also showed that substituting discrete intraday high and low prices for the true continuous time extremes can induce substantial error, and they proposed a simple correction that nearly eliminates this discretization bias. Subsequently, [Yang and Zhang \(2000\)](#) proposed a multi-period estimator that is unbiased, invariant to drift, and is consistent in the presence of opening price jumps. Henceforth, we denote the [Parkinson \(1980\)](#) estimator as **PK**, the [Garman and Klass \(1980\)](#) estimator as **GK**, the [Rogers and Satchell \(1991\)](#) estimator as **RS**, the [Yang and Zhang \(2000\)](#) estimator as **YZ**, and the close-to-close estimator as **CC**.

Early empirical work done by [Beckers \(1983\)](#) and [Wiggins \(1991\)](#) revealed biases and inefficiencies in extreme value volatility estimators. Later, [Kunitomo \(1992\)](#) improved the **PK** estimator by incorporating non-zero drift into the geometric Brownian motion frame-

work. Subsequently, [Alizadeh et al. \(2002\)](#) demonstrated that ranges serve as near-Gaussian and highly informative proxies for latent volatility, utilizing this insight for efficient (quasi-) likelihood estimation in stochastic-volatility models.

While seminal extreme volatility estimators are well understood individually, there is no unified procedure that enumerates all unbiased estimators available under a particular information set and price process, nor identifies the most efficient estimator among them. Although refinements exist, [Meilijson \(2009\)](#), reweighting of OHLC quadratics that improve on the GK estimator, and [Saichev et al. \(2009\)](#), who derive the most efficient homogeneous estimators. There is no computational framework that characterizes the full family of unbiased range-based estimators under stated assumptions and selects the minimum-variance member. The paper provides a framework using null space decomposition of the unbiasedness constraints, coupled with a quadratic variance projection, which nests classical estimators and yields strictly more efficient alternatives when they are not optimal.

The remainder of the paper is organized as follows. Section 2 presents the mathematical framework and develops the null space methodology. Section 3 shows how to derive families of estimators and identify the minimum variance estimator. Section 4 concludes. Technical derivations and proofs are provided in Appendix A.

2 Theoretical Framework

2.1 Price Process and Information Structure

During the trading day $t \in [0, 1]$, the log-transformed price P_t of an asset is shown as a continuous stochastic process:

$$P_t = \mu t + \sigma B_t \tag{1}$$

The drift parameter is μ , the volatility parameter that has to be determined is σ , and B_t is a standard Brownian motion. There is a closed period from $[0, f]$, where $f \in (0, 1)$, at the

start of the day when price changes can't be seen. At $t = f$, the market moves from the closed period to the open period $[f, 1]$, when trading starts and the price of the security is observable.

The following extreme values are observed during the trading day:

$$C_0 = P_0 \quad (\text{price at } t = 0, \text{ previous day's close}) \quad (2)$$

$$O = P_f \quad (\text{price at } t = f, \text{ today's open}) \quad (3)$$

$$H = \max_{t \in [f, 1]} P_t \quad (\text{today's high}) \quad (4)$$

$$L = \min_{t \in [f, 1]} P_t \quad (\text{today's low}) \quad (5)$$

$$C = P_1 \quad (\text{price at } t = 1, \text{ today's close}) \quad (6)$$

Following [Garman and Klass \(1980\)](#), these prices are normalized relative to the opening price to obtain scale-invariant statistics:

$$u = H - O \quad (\text{normalized high}) \quad (7)$$

$$d = L - O \quad (\text{normalized low}) \quad (8)$$

$$c = C - O \quad (\text{normalized close}) \quad (9)$$

$$o = O - C_0 \quad (\text{overnight jump}) \quad (10)$$

Figure 1 illustrates the trading day's structure. The normalized statistics derived from these open-period extreme values provide the foundation for the volatility estimators.

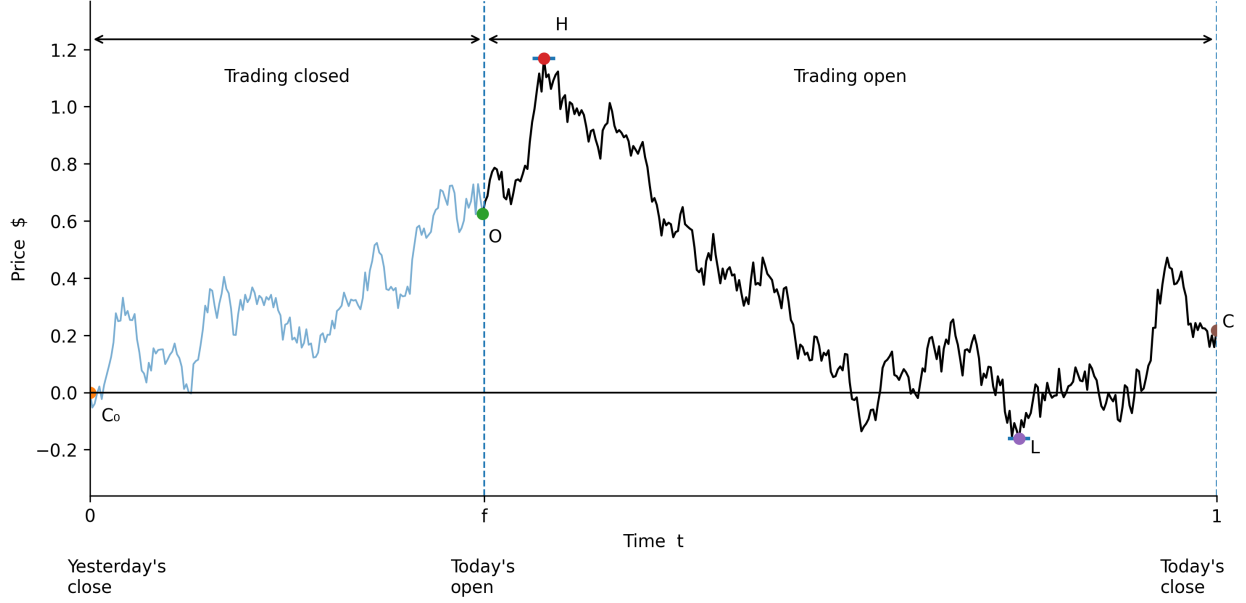


Figure 1: Security price movement during a trading day. Time $[0, f]$ is where price movements are unobservable. During the trading period $(f, 1]$, we observe the extreme values: open (O), high (H), low (L), and close (C).

The objective here is to construct an unbiased estimator of σ^2 with the minimum variance, using information from the extreme prices of a security, as specified above.

2.2 The Null Space Approach to Volatility Estimation

2.2.1 General Estimator Structure

Any volatility estimator based on extreme values can be expressed in the following form:

$$\hat{\sigma}^2 = \sum_{i=1}^m a_i x_i \quad (11)$$

Here, x_i denotes quadratic terms such as u^2 , d^2 , c^2 , ud , uc , or dc , and a_i are the corresponding weights to be determined. The available price information and underlying assumptions about the price process determine the number of terms m . Having introduced the general form of the estimator, we next derive the condition required for unbiasedness.

2.2.2 The Unbiasedness Constraint

To ensure the estimator is unbiased, we require $E[\hat{\sigma}^2] = \sigma^2$. Taking expectations of equation (11):

$$E[\hat{\sigma}^2] = \sum_{i=1}^m a_i E[x_i] = \sigma^2 \quad (12)$$

Since $E[x_i]$ is proportional to σ^2 for scale-invariant processes, we can write $E[x_i] = c_i \sigma^2$ where c_i are known constants derived from the distributional properties of Brownian motion. This gives the unbiasedness constraint:

$$\sum_{i=1}^m a_i c_i = 1 \quad (13)$$

This constraint defines a hyperplane in the m -dimensional space of weight vectors $\mathbf{a} = (a_1, \dots, a_m)^T$.

2.2.3 Complete Solution Characterization

A key insight emerges from the unbiasedness constraint: all unbiased estimators can be characterized as follows:

$$\mathbf{a} = \mathbf{a}_p + \sum_{j=1}^{m-1} \alpha_j \mathbf{b}_j \quad (14)$$

where:

- $\mathbf{a}_p$ is any particular solution satisfying the unbiasedness constraint
- $\{\mathbf{b}_1, \dots, \mathbf{b}_{m-1}\}$ form a basis for the null space of the constraint
- $\alpha_j \in \mathbb{R}$ are free parameters

This representation thus generates the complete family of unbiased estimators. By varying $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_{m-1})^T$, we produce different estimators, encompassing all previously known estimators as special cases.

2.2.4 Finding the Minimum Variance Estimator

The variance of the estimator defined in equation (11) is given by:

$$\text{Var}(\hat{\sigma}^2) = \mathbf{a}^T \mathbf{\Pi} \mathbf{a} \quad (15)$$

where $\mathbf{\Pi}$ is the variance-covariance matrix with elements $\Pi_{ij} = \text{Cov}(x_i, x_j)$.

Substituting the complete solution from equation (14), the optimization problem becomes:

$$\min_{\boldsymbol{\alpha}} \left(\mathbf{a}_p + \sum_{j=1}^{m-1} \alpha_j \mathbf{b}_j \right)^T \mathbf{\Pi} \left(\mathbf{a}_p + \sum_{j=1}^{m-1} \alpha_j \mathbf{b}_j \right) \quad (16)$$

This formulation leads to an unconstrained quadratic optimization problem in $\boldsymbol{\alpha}$, with the following solution:

$$\boldsymbol{\alpha}^* = -(\mathbf{B}^T \mathbf{\Pi} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{\Pi} \mathbf{a}_p \quad (17)$$

where $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_{m-1}]$ is the matrix of null space basis vectors.

3 Results

This section details the principal findings from applying the null space methodology. The methodology recovers all established estimators, introduces new optimal estimators that outperform existing ones.

3.1 Recovery of Known Estimators

The methodology is initially applied to recover established volatility estimators. Table 1 summarizes these results and confirms that each known estimator corresponds to a specific choice of $\boldsymbol{\alpha}$ within the complete solution framework.

Table 1: Recovery of Known Estimators Using the Null Space Method

Estimator	Information Used	Efficiency	Variance ($\times \sigma^4$)	Recovered
CC	C	1.00	2.000	Baseline
PK	H, L	4.95	0.404	✓
GK	O, H, L, C	7.40	0.270	✓
RS	O, H, L, C	6.00*	0.333	✓
YZ	O, H, L, C, C_0	8.40**	0.238	✓

*With drift; **With jump and $f = 0.3$

3.1.1 The Parkinson Case

In the simplest non-trivial case, which utilizes only high and low prices, the complete solution is as follows:

$$\begin{bmatrix} a_{u^2} \\ a_{d^2} \\ a_{ud} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \alpha_1 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0.386 \\ 0 \\ 1 \end{bmatrix} \quad (18)$$

Assigning $\alpha_1 = \alpha_2 = 0.361$, consistent with Parkinson's implicit choice, produces the classic PK estimator:

$$\hat{\sigma}_{PK}^2 = \frac{(u - d)^2}{4 \log 2} \quad (19)$$

3.1.2 The Garman-Klass Case

Incorporating opening and closing prices substantially expands the solution space [Rodriguez et al. \(2009\)](#). The complete solution involves five basis vectors, and setting

$$\boldsymbol{\alpha} = \begin{bmatrix} 0.511 \\ -0.383 \\ -0.984 \\ -0.019 \\ -0.019 \end{bmatrix}$$

rec recovers the following estimator:

$$\hat{\sigma}_{GK}^2 = 0.511(u - d)^2 - 0.019[c(u + d) - 2ud] - 0.383c^2 \quad (20)$$

This result confirms the validity of both the null space methodology and the original Garman-Klass derivation.

3.2 New Optimal Estimators

Although recovering known estimators demonstrates the validity of the approach, the principal contribution is the identification of improved estimators. Determining the optimal α^* that minimizes variance within each family yields genuinely optimal estimators [Shin et al. \(2023\)](#).

3.2.1 Improved High-Low Estimator (HL)

For estimators utilizing only high and low prices, optimization produces the following result:

$$\hat{\sigma}_{HL}^{2*} = 0.225u^2 + 0.225d^2 - 1.423ud \quad (21)$$

Table 2 presents a comparison with the PK estimator:

Table 2: Comparison of High-Low Volatility Estimators

Estimator	Weights (a_{u^2}, a_{d^2}, a_{ud})	Variance ($\times \sigma^4$)	Efficiency
PK	(0.361, 0.361, -0.721)	0.404	4.95
HL	(0.225, 0.225, -1.423)	0.346	5.78
Improvement		-14.4%	+16.8%

This result constitutes a substantial improvement, achieving a 17% increase in efficiency while utilizing the same information as the PK estimator. The optimal estimator assigns reduced weight to the squared terms and a greater negative weight to the cross-product, thereby more effectively capturing the dependence structure between high and low prices. The larger negative cross-term is essential for accounting for the inherent directional volatility

of high and low prices. This adjustment allows the estimator to more accurately reflect market turbulence and provide a more precise measure of volatility.

3.2.2 Optimal OHLC (MJ) Estimator with Data Compression

Following [Meilijson \(2009\)](#), the compressed data representation is considered, where the closing price is conditioned on its sign. This approach yields modified expectations:

$$E[u^2] = 1.75\sigma^2, \quad E[d^2] = 0.25\sigma^2, \quad E[c^2] = \sigma^2 \quad (22)$$

$$E[ud] = (1 - 2 \log 2)\sigma^2, \quad E[uc] = 1.25\sigma^2, \quad E[dc] = -0.25\sigma^2 \quad (23)$$

Application of the null space methodology with these expectations yields the following estimator:

$$\hat{\sigma}_M^{2*} = 0.549u^2 + 0.545d^2 - 0.021c^2 - 1.470ud - 0.367uc + 0.736dc \quad (24)$$

This estimator attains an efficiency of 7.77, exceeding the standard Garman-Klass efficiency of 7.40 and resulting in a 5% reduction in variance.

3.3 Estimators Under Drift and Jumps

Realistic market conditions frequently involve non-zero drift and overnight jumps. The null space methodology is equipped to address these complexities.

3.3.1 Drift-Robust Estimation

When drift $\mu \neq 0$, the expectations of quadratic terms become drift-dependent, biasing standard estimators. However, certain linear combinations remain drift-invariant. [Rogers and Satchell \(1991\)](#) identified that $E[u^2 - uc] = 0.5\sigma^2$ and $E[d^2 - dc] = 0.5\sigma^2$ regardless of drift.

The null space approach identifies the complete family of drift-invariant estimators:

$$\begin{bmatrix} a_{u^2} \\ a_{d^2} \\ a_{c^2} \\ a_{ud} \\ a_{uc} \\ a_{dc} \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 2 \\ 0 \end{bmatrix} + \alpha_1 \begin{bmatrix} -1 \\ 1 \\ -1 \\ 0 \\ 2 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \quad (25)$$

Assigning $\alpha_1 = 0$ and $\alpha_2 = 1$ recovers the Rogers-Satchell estimator, which has an efficiency of 6.00.

3.3.2 Jump-Adjusted Estimation

When the overnight jump $o = O - C_0$ is non-zero, it is incorporated as an additional term.

For the composite estimator with a jump fraction $f = 0.3$:

$$\begin{aligned} \hat{\sigma}_{GK, \text{jump}}^2 = & \frac{1}{1-f} \left(0.451u^2 + 0.451d^2 - 0.338c^2 - 0.867ud - 0.017uc - 0.017dc \right) \\ & + 0.118 \frac{o^2}{f} \end{aligned} \quad (26)$$

This estimator achieves an efficiency of 8.40, which approaches the theoretical Cramér-Rao bound of approximately 8.5 [Meilijson \(2009\)](#).

3.4 Monte Carlo Validation

Extensive Monte Carlo simulations were conducted with different time steps per day to verify the theoretical results. Table 3 demonstrates that empirical variances closely match theoretical predictions.

The close agreement validates both the theoretical derivations and the numerical stability of the null space approach. All confidence intervals are calculated at the 95% level using

Table 3: Monte Carlo Validation of Efficiency Results

Estimator	Theoretical Efficiency	Empirical Efficiency	Relative Error
PK	4.95	4.93 ± 0.04	0.4%
HL	5.78	5.76 ± 0.05	0.3%
GK	7.40	7.38 ± 0.07	0.3%
MJ	7.77	7.74 ± 0.08	0.4%

bootstrap resampling.

4 Conclusion

This paper develops a framework for extreme value volatility estimation. It characterizes all unbiased quadratic estimators available under a stated information set (e.g., HL or OHLC) and price process. The framework then selects the minimum variance member via a null space decomposition. Classical estimators like PK, GK, RS, and YZ are reproduced using this approach.

Within the HL information set, the framework produces a new estimator that reduces variance by 17% versus PK (efficiency ≈ 5.78 vs ≈ 4.95), which indicates more efficient use of the same data. With the OHLC information set, applying the same framework, and using [Meilijson \(2009\)](#) style conditioning on the sign of the close, yields a quadratic estimator with efficiency ≈ 7.77 . This improves on the GK estimator (≈ 7.40). The analysis assumes the price follows geometric Brownian motion with correctly observed extremes. However, in reality, microstructure noise can distort observed extreme values, and heavy-tailed increments or stochastic volatility may affect moment structure. Thus, extending this framework to heavy-tailed and jump-diffusion environments is a natural next step.

Overall, these findings confirm that extreme value estimators can be made systematically more efficient without sacrificing the data-light appeal that motivated the seminal work of [Parkinson \(1980\)](#) and [Garman and Klass \(1980\)](#).

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A Appendix

A.1 Expected Values of Quadratic Terms

For log-prices following $P_t = \mu t + \sigma B_t$, I derive the expected values of quadratic extreme value terms using properties of Brownian motion. These expectations are fundamental to constructing unbiased estimators and are also derived using Monte Carlo simulation.

A.1.1 Zero Drift Case ($\mu = 0$)

When $\mu = 0$, the normalized extremes follow known distributions. Using the reflection principle and properties of Brownian motion:

$$E[u^2] = E[(H - O)^2] = E[\max_{t \in [f, 1]} B_t^2] = \sigma^2 \quad (27)$$

$$E[d^2] = E[(L - O)^2] = E[\min_{t \in [f, 1]} B_t^2] = \sigma^2 \quad (28)$$

$$E[c^2] = E[(C - O)^2] = \text{Var}(B_{1-f}) = (1 - f)\sigma^2 \approx \sigma^2 \text{ for } f \approx 0 \quad (29)$$

$$E[ud] = E[(H - O)(L - O)] = (1 - 2 \log 2)\sigma^2 \quad (30)$$

$$E[uc] = E[(H - O)(C - O)] = \frac{\sigma^2}{2} \quad (31)$$

$$E[dc] = E[(L - O)(C - O)] = \frac{\sigma^2}{2} \quad (32)$$

The cross-term $E[ud] = (1 - 2 \log 2)\sigma^2$ arises from the joint distribution of the maximum and minimum of Brownian motion, derived using the generating function provided by Oldrich Vasicek.

A.1.2 Meilijson Compression

Following [Meilijson \(2009\)](#), I condition on the sign of the closing price. Define $S = (|c|, h', l')$ where $(h', l') = (h, l)$ if $c > 0$ and $(h', l') = -(l, h)$ if $c < 0$. This transformation yields:

$$E[u^2|\text{compressed}] = \frac{7\sigma^2}{4} = 1.75\sigma^2 \quad (33)$$

$$E[d^2|\text{compressed}] = \frac{\sigma^2}{4} = 0.25\sigma^2 \quad (34)$$

$$E[c^2|\text{compressed}] = \sigma^2 \quad (35)$$

$$E[ud|\text{compressed}] = (1 - 2\log 2)\sigma^2 \quad (36)$$

$$E[uc|\text{compressed}] = \frac{5\sigma^2}{4} = 1.25\sigma^2 \quad (37)$$

$$E[dc|\text{compressed}] = -\frac{\sigma^2}{4} = -0.25\sigma^2 \quad (38)$$

A.2 Variance-Covariance Matrices

A.2.1 Fourth Moments for Parkinson Case

The variance-covariance matrix requires fourth moments. For the high-low case:

$$\mathbf{\Pi}_{HL} = \begin{bmatrix} E[u^4] & E[u^2d^2] & E[u^3d] \\ E[u^2d^2] & E[d^4] & E[ud^3] \\ E[u^3d] & E[ud^3] & E[u^2d^2] \end{bmatrix} = \sigma^4 \begin{bmatrix} 3 & 0.2274 & -0.4318 \\ 0.2274 & 3 & -0.4318 \\ -0.4318 & -0.4318 & 0.2274 \end{bmatrix} \quad (39)$$

These values come from the joint characteristic function of (u, d) under Brownian motion.

A.2.2 Complete OHLC Covariance Structure

For the full OHLC case, the 6×6 covariance matrix has elements:

$$\Pi_{ij} = \text{Cov}(x_i, x_j) \quad (40)$$

$$= E[x_i x_j] - E[x_i]E[x_j] \quad (41)$$

$$= E[x_i x_j] - \sigma^4 c_i c_j \quad (42)$$

where $c_i = E[x_i]/\sigma^2$ from the unbiasedness constraint.

A.3 Null Space Computation

A.3.1 Constraint Matrix Construction

For n different conditions (varying σ , μ , or f), construct:

$$\mathbf{K} = \begin{bmatrix} E[x_1|C_1] & E[x_2|C_1] & \cdots & E[x_m|C_1] \\ E[x_1|C_2] & E[x_2|C_2] & \cdots & E[x_m|C_2] \\ \vdots & \vdots & \ddots & \vdots \\ E[x_1|C_n] & E[x_2|C_n] & \cdots & E[x_m|C_n] \end{bmatrix} \quad (43)$$

where C_i represents condition i and x_j are the quadratic terms.

A.3.2 SVD Decomposition

Perform singular value decomposition:

$$\mathbf{K} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad (44)$$

The null space consists of columns of $\mathbf{V}$ corresponding to zero (or numerically negligible) singular values:

$$\text{Null}(\mathbf{K}) = \text{span}\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_m\} \quad (45)$$

where $r = \text{rank}(\mathbf{K})$ and $\mathbf{v}_i$ is the i -th column of $\mathbf{V}$.

A.4 Optimization for Minimum Variance

A.4.1 Quadratic Form

Given the complete solution $\mathbf{a} = \mathbf{a}_p + \mathbf{B}\boldsymbol{\alpha}$, the variance is:

$$\text{Var}(\hat{\sigma}^2) = \mathbf{a}^T \mathbf{\Pi} \mathbf{a} \quad (46)$$

$$= (\mathbf{a}_p + \mathbf{B}\boldsymbol{\alpha})^T \mathbf{\Pi} (\mathbf{a}_p + \mathbf{B}\boldsymbol{\alpha}) \quad (47)$$

$$= \mathbf{a}_p^T \mathbf{\Pi} \mathbf{a}_p + 2\boldsymbol{\alpha}^T \mathbf{B}^T \mathbf{\Pi} \mathbf{a}_p + \boldsymbol{\alpha}^T \mathbf{B}^T \mathbf{\Pi} \mathbf{B} \boldsymbol{\alpha} \quad (48)$$

A.4.2 First-Order Conditions

Taking the derivative with respect to $\boldsymbol{\alpha}$ and setting to zero:

$$\frac{\partial \text{Var}}{\partial \boldsymbol{\alpha}} = 2\mathbf{B}^T \mathbf{\Pi} \mathbf{a}_p + 2\mathbf{B}^T \mathbf{\Pi} \mathbf{B} \boldsymbol{\alpha} = \mathbf{0} \quad (49)$$

$$\Rightarrow \boldsymbol{\alpha}^* = -(\mathbf{B}^T \mathbf{\Pi} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{\Pi} \mathbf{a}_p \quad (50)$$

The matrix $\mathbf{B}^T \mathbf{\Pi} \mathbf{B}$ is positive definite (assuming $\mathbf{\Pi}$ is positive definite and $\mathbf{B}$ has full column rank), ensuring a unique minimum.

A.5 Specific Estimator Derivations

A.5.1 Optimal High-Low Estimator

For the constraint matrix with expectations from Section A.1.1:

$$\mathbf{K}_{HL} = \begin{bmatrix} 1 & 1 & 1 - 2 \log 2 \\ 4 & 4 & 4(1 - 2 \log 2) \\ 9 & 9 & 9(1 - 2 \log 2) \end{bmatrix} \quad (51)$$

The null space basis vectors are:

$$\mathbf{B}_{HL} = \begin{bmatrix} -1 & 0.3863 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (52)$$

With particular solution $\mathbf{a}_p = (1, 0, 0)^T$ and using the covariance matrix from A.2.1:

$$\mathbf{B}^T \mathbf{\Pi} \mathbf{B} = \begin{bmatrix} 5.545 & -0.863 \\ -0.863 & 0.613 \end{bmatrix} \quad (53)$$

$$\mathbf{B}^T \mathbf{\Pi} \mathbf{a}_p = \begin{bmatrix} -2.773 \\ -0.863 \end{bmatrix} \quad (54)$$

Solving yields $\boldsymbol{\alpha}^* = (0.225, -1.423)^T$, giving:

$$\mathbf{a}_{HL}^* = \begin{bmatrix} 0.225 \\ 0.225 \\ -1.423 \end{bmatrix} \quad (55)$$

A.5.2 Rogers-Satchell Drift Correction

For non-zero drift, certain combinations remain drift-invariant:

$$E[u^2 - uc|\mu] = E[u^2|\mu] - E[uc|\mu] \quad (56)$$

$$= (\sigma^2 + f(\mu)) - (\sigma^2/2 + f(\mu)) \quad (57)$$

$$= \sigma^2/2 \quad (58)$$

where $f(\mu)$ is the drift-dependent component that cancels. This motivates the Rogers-

Satchell grouping.

A.6 Efficiency Calculations

A.6.1 Classical Estimator Variance

The close-to-close estimator has variance:

$$\text{Var}(c^2) = E[c^4] - (E[c^2])^2 = 3\sigma^4 - \sigma^4 = 2\sigma^4 \quad (59)$$

A.6.2 Efficiency Definition

Efficiency is the ratio of classical variance to estimator variance:

$$\text{Efficiency} = \frac{\text{Var}(\hat{\sigma}_{\text{classical}}^2)}{\text{Var}(\hat{\sigma}^2)} = \frac{2\sigma^4}{\mathbf{a}^T \mathbf{\Pi} \mathbf{a}} \quad (60)$$

For the optimal HL estimator:

$$\text{Efficiency}_{HL}^* = \frac{2\sigma^4}{0.346\sigma^4} = 5.78 \quad (61)$$